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example, in summer, the rate of the ascent of sap is more due to transpiration in spite of the fact that root pressure is very low. On the other hand, in winter when the rate of ascent of sap is low, a high root pressure is found.

iii. Ascent of sap continues even in the absence of roots

iv. The magnitude of root pressure is about 2atm, which can raise the water level up to few feet only, whereas the tallest trees are more than 100m high.

### 11.5.4 Physical Force Theory

Physical force theories suggest that ascent of sap takes place through the dead xylem vessel and the mechanism is entirely physical and living cells are not involved.

**1. Capillary theory**

**Boehm** (1809) suggested that the xylem vessels work like a capillary tube. This capillarity of the vessels under normal atmospheric pressure is responsible for the ascent of sap. This theory was rejected because the magnitude of capillary force can raise water level only up to a certain height. Further, the xylem vessels are broader than the tracheid which actually conducts more water and against the capillary theory.

**2. Imbibition theory**

This theory was first proposed by **Unger** (1876) and supported by **Sachs** (1878). This theory illustrates, that water is imbibed through the cell wall materials and not by the lumen. This theory was rejected based on the ringing experiment, which proved that water moves through the lumen of the cell and not by a cell wall.

**3. Cohesion-tension or Cohesion and transpiration pull theory**

Cohesion-tension theory was originally proposed by **Dixon** and **Jolly** (1894) and again put forward by **Dixon** (1914, 1924). This theory is based on the following features:

i. **Strong cohesive force or tensile strength of water**

Water molecules have the strong mutual force of attraction called **cohesive force** due to which they cannot be easily separated from one another. Further, the attraction between a water molecule and the wall of the xylem element is called **adhesion**. These cohesive and adhesive force works together to form an unbroken continuous water column in the xylem. The magnitude of the cohesive force is much high (350 atm) and is more than enough to ascent sap in the tallest trees.

ii. **Continuity of the water column in the plant**

An important factor which can break the water column is the introduction of air bubbles in the xylem. Gas bubbles expanding and displacing water within the xylem element is called **cavitation** or **embolism**. However, the overall continuity of the water column remains undisturbed since water diffuses into the adjacent xylem elements for continuing ascent of sap.

iii. **Transpiration pull or Tension in the unbroken water column**

The unbroken water column from leaf to root is just like a rope. If the rope is pulled from the top, the entire rope will move upward. In plants, such a pull is generated by the process of transpiration which is known as **transpiration pull.**